

CÓDIGO:

Question 13. Water in the Swiss Alps (15% of the total; 11th and 12th grades)

13.1	13.2	13.3	13.4	13.5	13.6	13.7	13.8	13.9	13.10	Total	Total
5	10	4	3	6	6	8	10	6	2	60	15%

When Erik and César went to the International Chemistry Olympiad last July, they discovered glaciers on Mount Säntis, more than two kilometers above sea level. To their surprise, the ice remained intact during the mentors' visit to the mountain, even though the temperature was slightly above 0 °C. Intrigued by this discovery, they decided to dust off Peter Atkins' old Physical Chemistry book that was abandoned on the bookshelf and scrutinize its pages in search of answers.

First, they found that atmospheric pressure decreases with increasing height h with respect to sea level according to the barometric equation. In the case of the troposphere, where the temperature varies approximately linearly with a ratio $L_b = 6.5 \text{ K} / 1000 \text{ m}$.

Variation of pressure with height in the troposphere

$$P' = P_b \cdot \left[\frac{T_b - h \cdot L_b}{T_b} \right]^{\frac{g^\circ \cdot M}{R \cdot L_b}}$$

where P is the pressure at height h , above sea level; $P_b = 101.325 \text{ kPa}$ and $T_b = 288.15 \text{ K}$ are the reference pressure and temperature, respectively; $g^\circ = 9.80665 \text{ m/s}^2$ is the acceleration of Earth's gravity; $M = 0.0289644 \text{ kg/mol}$ is the average molecular mass of air; and R is the gas constant.

13.1. Calculate the height (in m) of Mount Säntis if the barometric pressure P' at the summit is 74.7 kPa.

Then, Erik came up with an equation that allows us to calculate the phase transition temperature when the pressure differs from the standard pressure.

Clausius-Clapeyron' equation

$$\frac{dP}{dT} = \frac{\Delta S_{\text{trans}}}{\Delta V_{\text{trans}}}$$

where dP/dT represents the slope of the pressure vs temperature curve when two phases coexist in equilibrium; ΔS_{trans} and ΔV_{trans} are the entropy and volume changes for the phase transition, respectively.

For cases of change of state of aggregation of a pure substance, considering that ΔS_{trans} and ΔV_{trans} are approximately constant in a small temperature range and if the vapor in equilibrium behaves like an ideal gas, the previous equation is transformed to:

Change of solid-liquid state: $\frac{P - P^\circ}{T_{\text{fus}} - T^\circ_{\text{fus}}} = \frac{\Delta H^\circ_{\text{fus}}}{\Delta V^\circ_{\text{fus}}}$ Change of liquid-vapor state: $\ln\left(\frac{P}{P^\circ}\right) = -\frac{\Delta H^\circ_{\text{vap}}}{R} \cdot \left(\frac{1}{T_{\text{eb}}} - \frac{1}{T^\circ_{\text{eb}}}\right)$

where $T^\circ_{\text{fus}} = 273,15 \text{ K}$ and $T^\circ_{\text{eb}} = 373,15 \text{ K}$ are the melting and boiling temperatures at standard pressure $P^\circ = 1,00 \text{ bar} = 100 \text{ kPa}$, respectively; $\Delta H^\circ_{\text{fus}}$ and $\Delta H^\circ_{\text{vap}}$ are the standard enthalpies of fusion and vaporization; $\Delta V^\circ_{\text{fus}}$ is the volume change that occurs when fusion occurs.

	$\Delta H^\circ_f / \text{kJ} \cdot \text{mol}^{-1}$	$S^\circ / \text{J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	$\rho / \text{g} \cdot \text{dm}^{-3}$
H ₂ O(s)	-291,8 (25 °C)	43,2 (0 °C)	917
H ₂ O(l)	-285,8 (25 °C)	65,2 (0 °C) 69,9 (25 °C) 86,8 (100 °C)	1000
H ₂ O(g)	-241,8 (25 °C)	196,9 (100 °C)	0,598

CÓDIGO:

13.2. Calculate $\Delta H^\circ_{\text{fus}}$, $\Delta S^\circ_{\text{fus}}$ (in J/(mol·K)), $\Delta V^\circ_{\text{fus}}$ (in dm³/mol), $\Delta H^\circ_{\text{vap}}$ and $\Delta S^\circ_{\text{vap}}$ (in J/(mol·K)) for water from the thermodynamic data above.

13.3. Using the above results and the Clausius-Clapeyron equations for state changes, calculate the melting temperatures T_{fus} and boiling temperatures T_{eb} (in °C) of water at 74.7 kPa.

The previous results, therefore, do not justify the existence of ice under these conditions. The students became more thoughtful when they noticed that the weather station on top of the mountain indicated 100 kPa, 5.0 °C and a relative humidity of 80%.

13.4. To be sure of all his reasoning, Caesar calculated the spontaneity of the ice-melting process under these conditions: what result did he obtain?

You must calculate $\Delta G^\circ_{\text{fus}}$ (5 °C) in kJ/mol.

Then it occurred to Erik that perhaps there was no water in the mountains because it was evaporating. Just at that moment, a strong wind began to blow, the clouds loaded with moisture collided with the mountains, and it started to rain, even from the bottom up, a product of the orography. He realized that for condensation to occur at a given temperature, the air must first be saturated with water vapor; the relative humidity must be $H_r = 100\%$ inside the cloud.

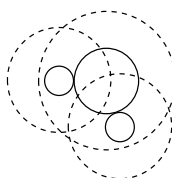
13.5. Calculate the pressure of water vapor at 5.0 °C when the air is saturated with moisture $P_{\text{H}_2\text{O}}^{\text{sat}}$ in kPa.

13.6. Help Erik calculate how much water gas (in g) is contained in 10 km³ of air around Mount Sántis when the relative humidity is 80% at 5.0°C. Consider $P_{\text{H}_2\text{O}}^{\text{sat}} = 0.50$ kPa if he failed to solve **13.5**.

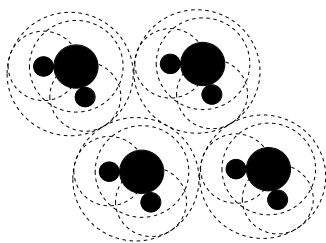
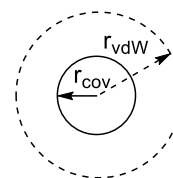
13.7. Condensation also requires a nucleus or surface where the molecules that “escape” from the gas phase are deposited. How much water per unit area is contained in a monolayer (layer one molecule thick) covering the condensation nucleus? Assume that the water molecules are spheres of radius $r = 179$ pm and are arranged as compactly as possible.

Effective radius of a water molecule in constant motion

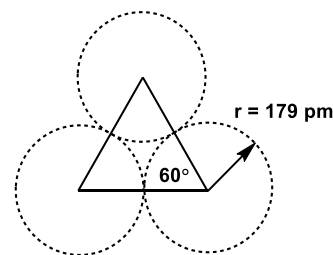
$$r_{\text{H}_2\text{O}}^{\text{eff}} \approx \frac{r_{\text{H}}^{\text{cov}} + r_{\text{H}}^{\text{vdW}} + r_{\text{O}}^{\text{cov}} + r_{\text{O}}^{\text{vdW}}}{2} = 179 \text{ pm}$$



Covalent and
van der Waals
radii



Arrangement of water molecules
in a compact monolayer



13.8. During his search, Erik found a curious fact in the chapter on chemical equilibrium: “In acidic aqueous solution, hydronium ions (H_3O^+), the Zundel cation (H_5O_2^+ , $\text{H}_3\text{O}^+ \cdot \text{H}_2\text{O}$, $\text{H}^+ \cdot 2\text{H}_2\text{O}$), the Eigen cation (H_9O_4^+ , $\text{H}_3\text{O}^+ \cdot 3\text{H}_2\text{O}$) and the Reed-Stoyanov cation ($\text{H}_{13}\text{O}_6^+$, $\text{H}^+ \cdot 6\text{H}_2\text{O}$) are present”. Although this did not help solve his questions, he represented the structures of these four species for fun, marking the relevant hydrogen bonds with dashed lines. Can you do it, too? Hint: All structures have some element of symmetry.

CÓDIGO:

13.9. César likes to experiment, and at lunchtime, he poured some of the contents of the salt shaker out the window. When he came out, he found that part of the ice outside had melted, exposing a portion of the ground. What temperature is required for water to freeze again? A saturated sodium chloride solution in water at 5.0 °C contains 26.3% by mass. $M(\text{NaCl}) = 58.44 \text{ g/mol}$

$$K_{\text{cryo}} = \frac{R \cdot M_{\text{dis}} \cdot T_{\text{fus}}^2}{1000 \cdot \Delta H_{\text{fus}}^{\circ}}$$

13.10. While reflecting, César remembered his visit to the Faculty of Chemistry and the debates with the professors of Thermodynamics and Kinetics, José Manuel Nieto and Rubén Álvarez and at that time, the mystery was solved! Can you infer why there is ice on Mount Sántis in summer?

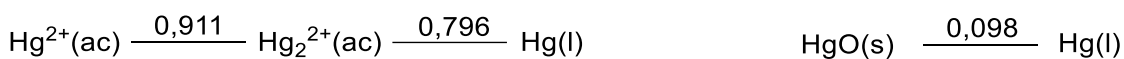
Question 14. Tasty electrochemistry! (15% of the total; 11th and 12th grades)

14.1	14.2	14.3	14.4	14.5	14.6	14.7	14.8	14.9	14.10	14.11	14.12	Total	Total
4	3	2	5	5	3	7	9	8	5	5	4	60	15%

One morning, the Ibero-American contestants Mario and Héctor studied various electrochemical systems. They had a silver chloride reference electrode (AgCl/Ag , $E^{\circ} = 0.222 \text{ V}$) and a digital multimeter. They decided to measure the potential of another electrode that had been prepared by the laboratory professor using mercury, mercurous sulfate, and a saturated solution of potassium sulfate since they could not find the value in the library books.

14.1. The electrochemical cell formed by the union of both electrodes had an EMF of 393 mV, with the silver chloride electrode acting as the anode. Represent the thermodynamic notation of the cell and calculate the unknown electrode potential $E[\text{Hg}_2\text{SO}_4(\text{s})/\text{Hg}(\text{l}), \text{SO}_4^{2-}(\text{ac})]$.

14.2. The presence of the mercurous ion caught Mario's attention, who believed that this ion was not stable in an aqueous solution. He demonstrates that the previous reasoning is incorrect, using the Latimer diagram of mercury.



14.3. Mario was convinced that the mercury(I) concentration had to be very small and that mercurous sulfate should be poorly soluble in water. How could you prove this hypothesis simply in the laboratory?

14.4. Héctor then found a note from the professor that the electrode had been prepared using a saturated solution of potassium sulfate, which has 10.0% salt by mass and a density of 1.08 g/cm³. How can the solution be prepared and what is its molar concentration? $M(\text{K}_2\text{SO}_4) = 174,27 \text{ g/mol}$.

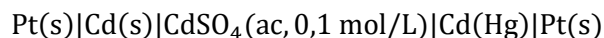
14.5. Then, Héctor decided to check the previous result. He poured a 5.00 mL aliquot of a freshly prepared saturated potassium sulfate solution into a beaker and added an excess concentrated barium chloride solution. After letting it rest, he filtered out all the white solid formed, dried it and determined that its mass was 720 mg. What is the new concentration value?

14.6. Mario observed that a tray with elemental sulfur had been placed on the work table. This is a standard procedure to collect accidentally spilled mercury residue. Because?

14.7. After examining all the data, Mario calculated the concentration of mercury(I) ions in the solution and the solubility product constant of mercurous sulfate. What values did he obtain? If I cannot solve **14.1**, consider that the potential of the mercury electrode was 0.620 V.

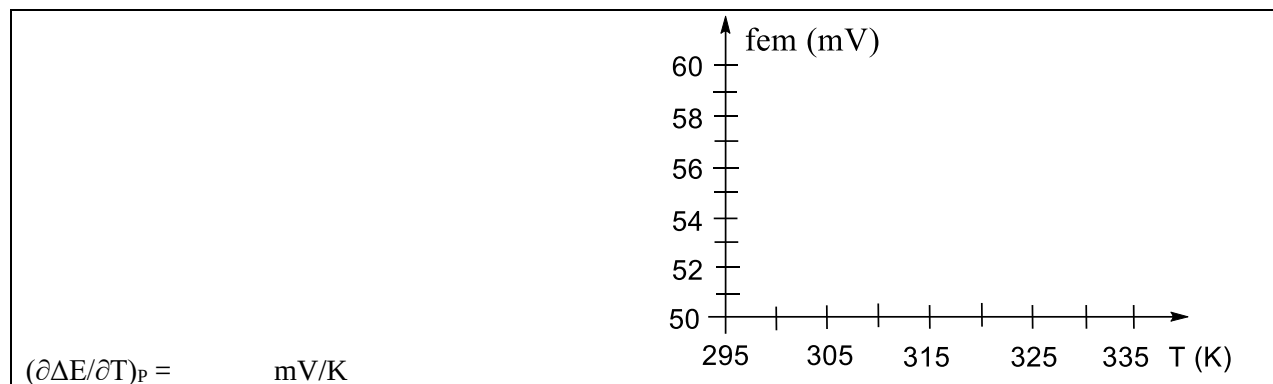
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There was also cadmium in the laboratory, and the students decided to study the formation reaction of an amalgam of this metal, for which they measured the EMF of a new cell.



T(°C)	23,5	30,3	34,6	39,2	44,0	48,6	54,9	60,4
FEM (mV)	52	53	54	55	56	57	59	60

14.8. Using the data provided, complete the ΔE vs T graph and calculate the cell temperature coefficient $(\partial\Delta E/\partial T)_P$ (the slope of the curve), assuming that the relationship between the variables is linear.



14.9. Calculate the value of the indicated thermodynamic variables at 298.15 K, with two significant figures.

$$\Delta S = -z \cdot F \cdot (\partial\Delta E/\partial T)_P$$

You must calculate ΔE (in mV), ΔG (in kJ/mol), ΔS (in J/(mol·K)) and ΔH (in kJ/mol). $T = 298,15 \text{ K}$

After work, the students drank carbonated soda. While they were resting, Héctor wondered why, if soda cans are made of aluminum, an active metal, no chemical reaction occurs. This question made them search the Internet, where they found the information they needed.

14.10. The typical acidulant in carbonated soft drinks is phosphoric acid. What is the acid concentration capable of giving the drink an acidity of $\text{pH} = 3.0$? $\text{pK}_{a1} = 2,15$; $\text{pK}_{a2} = 7,20$; $\text{pK}_{a3} = 12,35$

14.11. Show that the reaction between metallic aluminum and the acid in the soft drink is spontaneous at room temperature. $E^\circ(\text{Al}^{3+}/\text{Al}) = -1.66 \text{ V}$. You must calculate ΔE (in V) and ΔG (in kJ/mol)

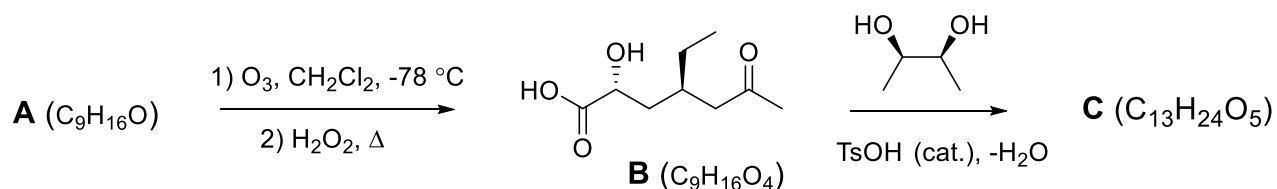
14.12. In reality, the aluminum in soda cans is covered with a layer of complex composition that protects it against oxidation due to acid solution. What maximum value can the dissolved aluminum concentration take within the layer? $\text{pK}_{ps}[\text{Al}(\text{OH})_3] = 32.97$.

Question 15. Play my chemistry in stereo! (12% of the total; only 12th grade)

15.1	15.2	15.3	15.4	15.5	15.6	15.7	15.8	15.9	15.10	Total	Total
6	2	5	5	4	4	4	9	3	6	48	12%

In the past, chemists widely used ozonolysis to determine the structure of organic compounds. This reaction also allows the formation of a doubly functionalized derivative starting from cyclic structures, such as chiral alcohol **A**, as shown in the following sequence.

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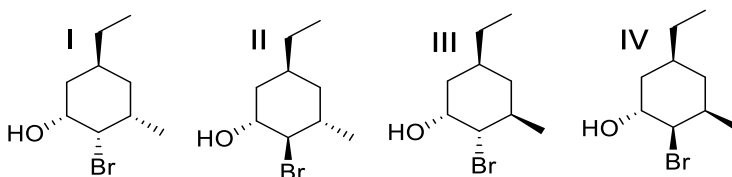
Throughout the question, you must correctly assign the stereochemistry of all centers.

15.1. Represent the structures of **A** and **C**. Tell how many stereoisomers each molecule has.

15.2. Name compound **B**.

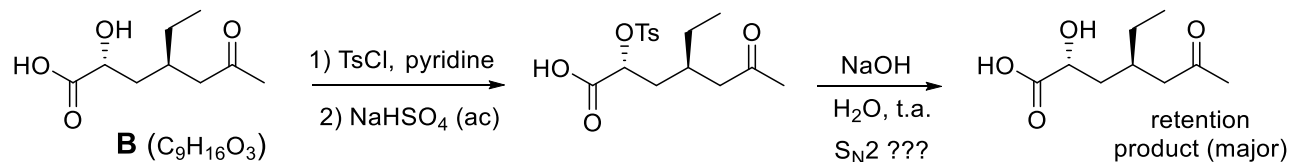
15.3. Represent the most important conformations of *meso*-2,3-butanediol (see structure in the diagram above) using Newman projections. Predict which conformation has the least energy and explain your reasoning.

15.4. What two stereoisomers have been shown to generate **A** by bimolecular elimination? Represent the equilibrium of their *chair* conformations and indicate the reactive conformation.



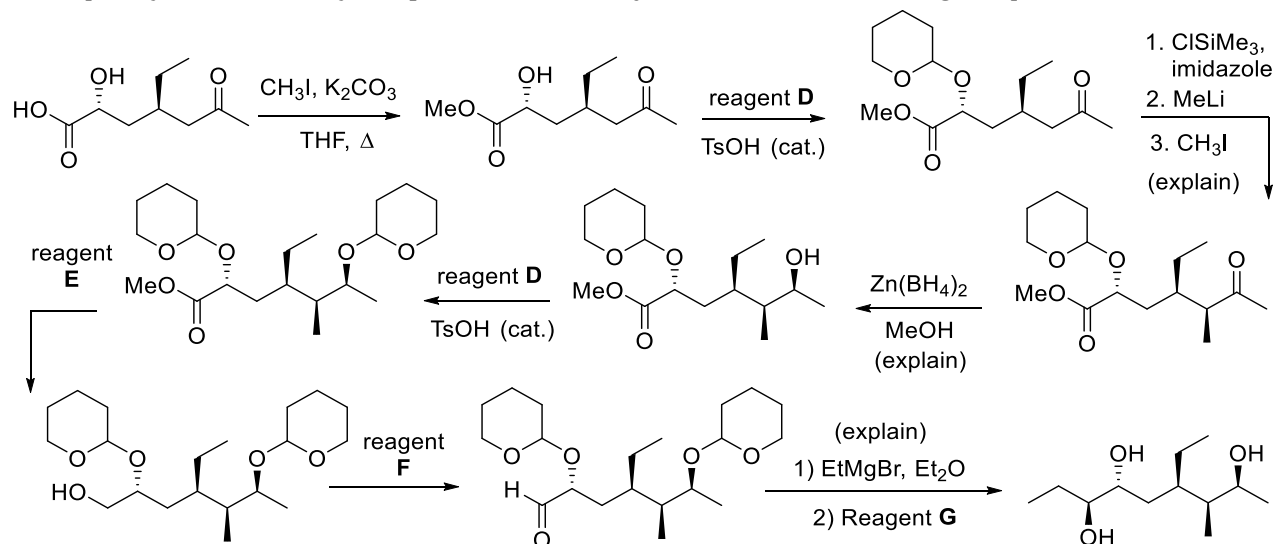
15.5. Ketol **B** exists in the form of two cyclic hemiacetals **B'** and **B''** that are diastereomers of each other. Represent the equilibrium that exists between the hemiketol *chair* conformations.

To change the stereochemistry of alcohol **B**, it was treated with tosyl chloride and then sodium hydroxide was added. However, when the product was analyzed, an appreciable amount of the starting was detected.



15.6. Pose a reaction mechanism that explains the result of the second step above.

Subsequently, the ozonolysis product was subjected to the following sequence of reactions.



CÓDIGO:

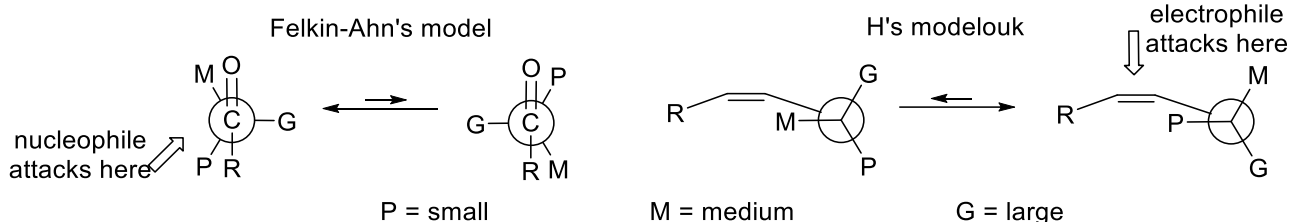
15.7. Suggest what the reagents could be **D**, **E**, **F** and **G**.

15.8. Propose mechanisms that justify the stereoselectivity of the indicated nucleophilic methylation, reduction and addition steps, using the Felkin-Ahn or Houk models, as appropriate. In your scheme, you only need to represent the conformation of the reactants that give rise to the product.

15.9. Represent the Fischer projection of the triol product of the above sequence. Classify the absolute stereochemistry of carbons according to Cahn-Ingold-Prelog.

15.10. Represent the major products of the reaction of compound **A** with: i) AMCPB, ii) $\text{CH}_2\text{I}_2/\text{Zn}(\text{Cu})$ y iii) $\text{Br}_2, \text{H}_2\text{O}$.

TsOH: *para*-toluenesulfonic acid; AMCPB: meta-chloroperoxy-benzoic acid; THF: tetrahydrofuran.



Question 16. Aromatic chemistry (13% of the total; only 12th grade)

16.1	16.2	16.3	16.4	16.5	16.6	16.7	16.8	16.9	16.10	16.11	16.12	16.13	16.14	Total	Total
5	3	2	1	6	1	5	1	7	4	7	3	4	3	52	13%

When an organic chemist observes the hexagons of wasp or bee honeycombs, he cannot help but think of the benzene ring, the aromatic organic substance par excellence. In this question, several synthetic sequences are shown that allow the obtaining of compounds of this type.

16.1. Fill in the missing reactants, structures, and reaction names in the scheme.

16.2. Compound **A** exists in the form of two other tautomers **A'** and **A''**, represent their structures and say why the tautomer shown is the most stable of all.

16.3. Propose a detailed mechanism for the hydrolysis reaction of **E** forming **F**.

16.4. Why does the isomerization reaction of *cis* **C** bicyclo occur in *trans* **D**?

16.5. Fill in the missing reactants, structures, and reaction names in the scheme. 16.6. Name compound **G**.

16.7. Why aluminum chloride that can act as a catalyst? Represent the electronic distribution of this molecule using hybrid orbitals for the central atom.

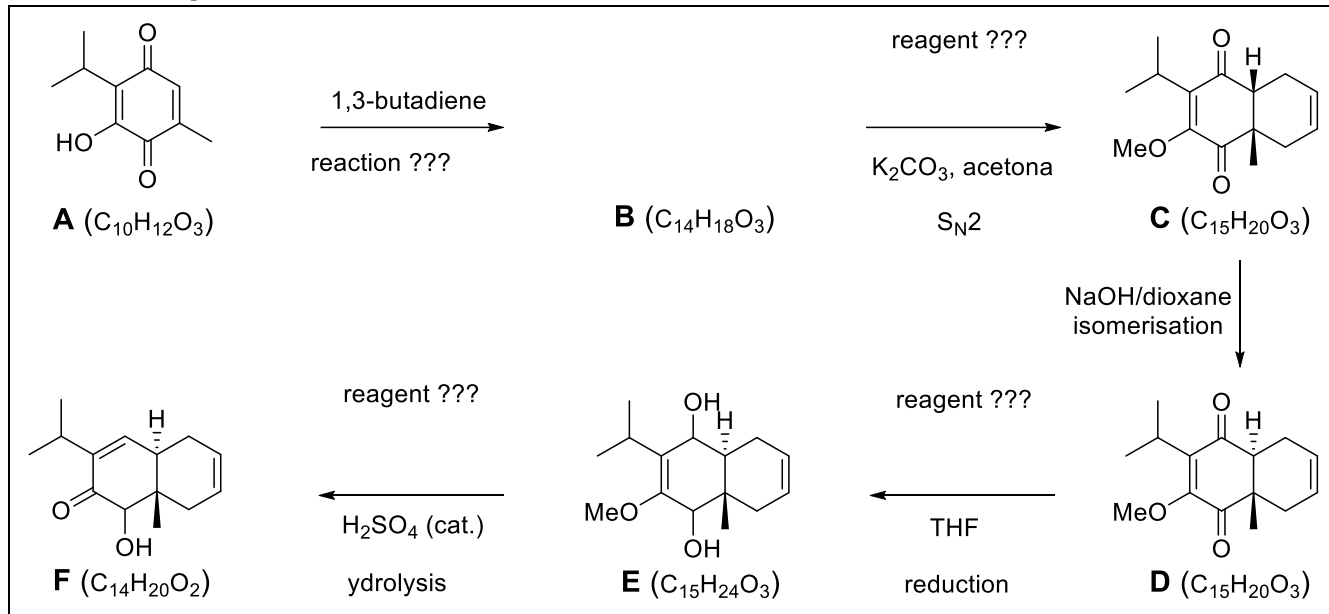
16.8. Compare the acidity of **G** and **M**. Explain your reasoning.

16.9. Fill in the missing reactants, structures, and reaction names in the scheme.

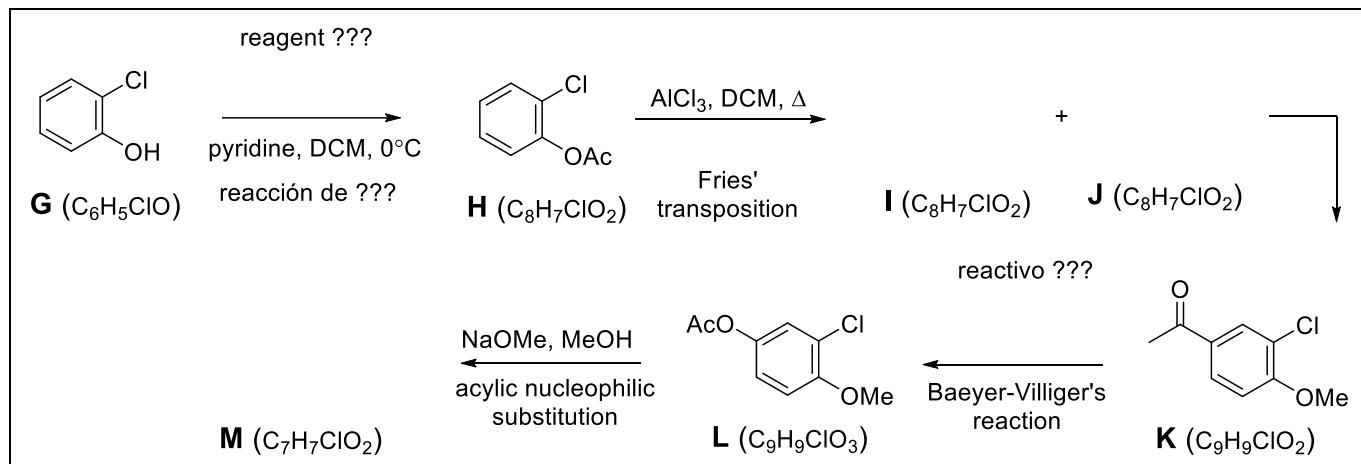
16.10. Propose mechanisms for Kornblum oxidation and cyanide-catalyzed rearrangement.

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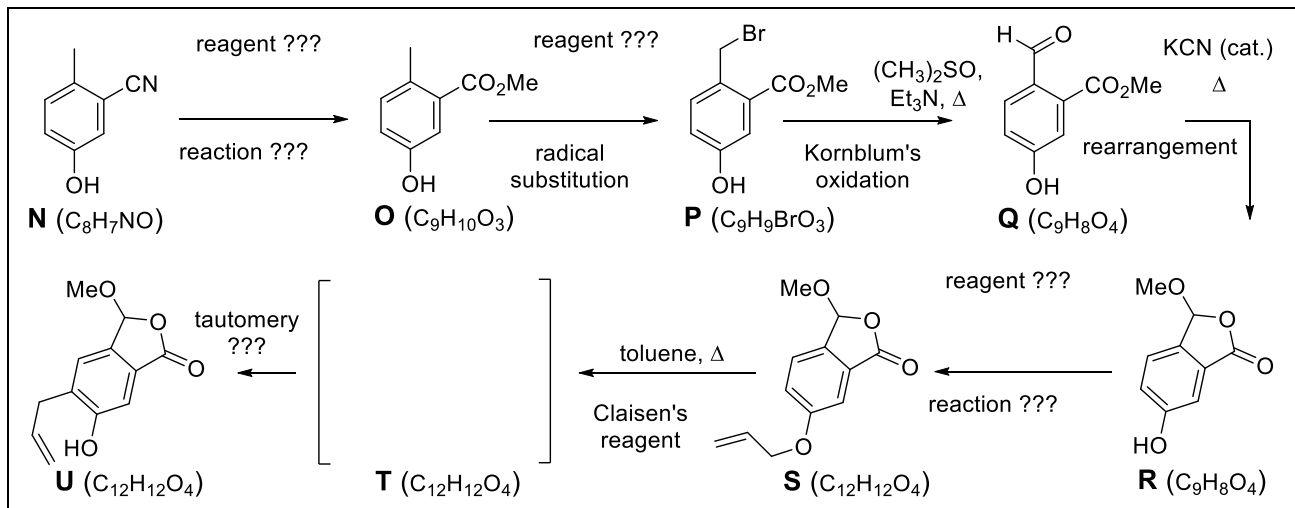
Scheme for question 16.1.



Scheme for question 16.5.

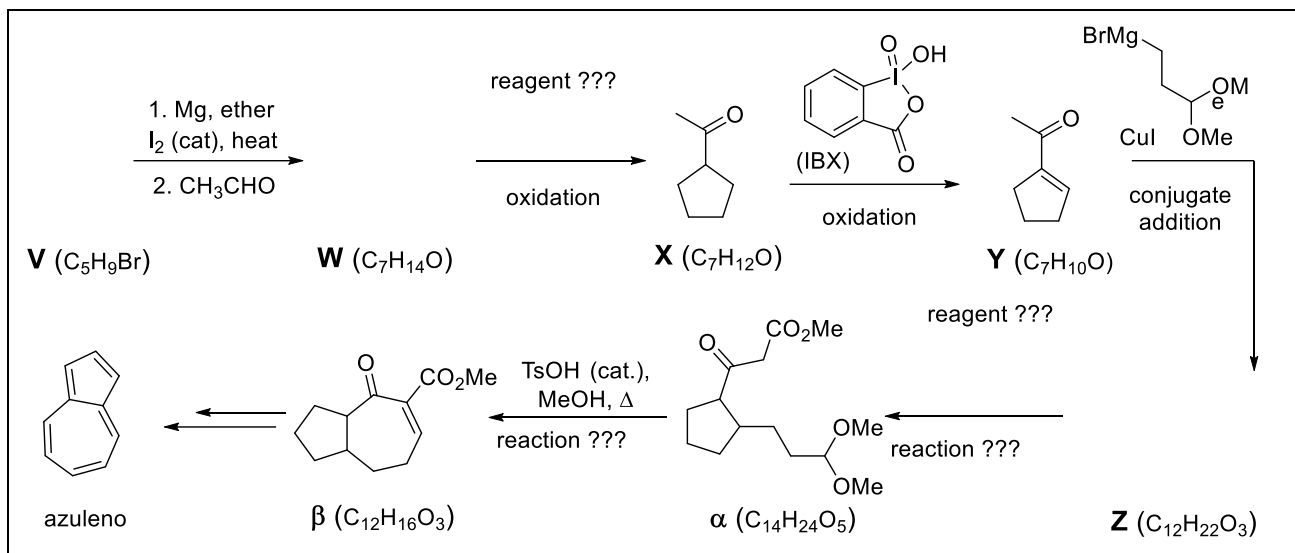


Scheme for question 16.9.



CÓDIGO:

16.11. Fill in the missing reactants, structures, and reaction names in the scheme.



16.12. Propose a mechanism for the ring closure reaction of **α** in an acidic medium.

16.13. Propose a sequence of reagents that can ultimately be used to obtain azulene.

16.14. Why is azulene aromatic and has high polarity? Represent four resonance structures.

Constante de Avogadro	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ecuación del gas ideal	$p \cdot V = n \cdot R \cdot T$
Constante de los gases	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	Primer Principio de la Termodinámica	$Q = \Delta U + W$
Cero grado Celsius	273,15 K	Trabajo	$W = p \cdot \Delta V = F \cdot d$
Constante de Faraday	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Energía de Gibbs	$\Delta G = \Delta H - T \cdot \Delta S$
Producto iónico del agua a 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Ecuación de Nernst	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \left(\frac{C_{\text{red}}}{C_{\text{ox}}} \right)$
Ecuación de Arrhenius	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$	$\Delta_r G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$	
Cinética de primer orden	$\ln c_t = \ln c_0 - k_r \cdot t$	Factor de van't Hoff	$i = 1 + \alpha \cdot (v - 1)$
	$t_{1/2} = \frac{\ln 2}{k_r}$	Crioscopía	$\Delta T_{\text{crio}} = i \cdot K_{\text{crio}} \cdot b$
Absorbancia	$A_\lambda = -\log_{10} \left(\frac{I}{I_0} \right) = \varepsilon \cdot l \cdot c$	Descenso de la presión de vapor	$\Delta p = \frac{i \cdot x_s}{x_{\text{dis}} + \sum i \cdot x_s} \cdot p^\circ_{\text{dis}}$
Volumen de una esfera	$V_{\text{sp}} = \frac{4}{3} \cdot \pi \cdot r^3$	Condiciones estándar	$p^\circ = 1 \text{ bar}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$
Área de una esfera	$A_{\text{sp}} = 4 \cdot \pi \cdot r^2$	Energía de un fotón	$E = h \cdot \nu = h \cdot \frac{c}{\lambda}$
Área de un círculo	$A_{\text{sp}} = \pi \cdot r^2$	Velocidad de la luz	$c = 2,998 \cdot 10^8 \text{ m} \cdot \text{s}^{-1}$
Longitud de una circunferencia	$l_c = 2 \cdot \pi \cdot r$	Constante de Planck	$h = 6,626 \cdot 10^{-34} \text{ J} \cdot \text{s}$
Fuerza de gravedad	$F_g = m \cdot g$	Aceleración de la gravedad en la Tierra	$g = 9,81 \text{ m} \cdot \text{s}^{-2}$
$1 \text{ J} = 1 \text{ C} \cdot \text{V} = 1 \text{ A} \cdot \text{s} \cdot \text{V} = 1 \text{ kPa} \cdot \text{L} = 1 \text{ Pa} \cdot \text{m}^3 = 1 \text{ N} \cdot \text{m} = 1 \text{ W} \cdot \text{s} = 0,2389 \text{ cal}$			
$1 \text{ atm} = 760 \text{ Torr} = 760 \text{ mm} = 101,325 \text{ kPa}$ $1 \text{ bar} = 100 \text{ kPa}$ $1 \text{ eV} = 96,485 \text{ kJ} \cdot \text{mol}^{-1}$			
$\pi = 3,1416$ $1 \text{ pm} = 0,01 \text{ \AA} = 10^{-12} \text{ m}$ $1 \text{ dm}^3 = 1 \text{ L}$ $1 \text{ mL} = 1 \text{ cm}^3$ $1 \text{ M} \equiv 1 \text{ mol/L}$			

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 H 1.008																	2 He 4.003
2	3 Li 6.941	4 Be 9.012											5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.180
3	11 Na 22.990	12 Mg 24.305											13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.066	17 Cl 35.453	18 Ar 39.948
4	19 K 39.098	20 Ca 40.078	21 Sc 44.956	22 Ti 47.867	23 V 50.942	24 Cr 51.996	25 Mn 54.938	26 Fe 55.845	27 Co 58.933	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	31 Ga 69.723	32 Ge 72.61	33 As 74.922	34 Se 78.96	35 Br 79.904	36 Kr 83.80
5	37 Rb 85.468	38 Sr 87.62	39 Y 88.906	40 Zr 91.224	41 Nb 92.906	42 Mo 95.94	43 Tc 98.906	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.905	54 Xe 131.29
6	55 Cs 132.91	56 Ba 137.33	57 La 138.91	* 72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po [209]	85 At [210]	86 Rn [222]
7	87 Fr [223]	88 Ra [226]	89 Ac [227]	** 104 Rf [265]	105 Db [268]	106 Sg [271]	107 Bh [270]	108 Hs [277]	109 Mt [276]	110 Ds [281]	111 Rg [280]	112 Cn [285]	113 Nh [284]	114 Fl [289]	115 Mc [288]	116 Lv [293]	117 Ts [294]	118 Og [294]
*	58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm [145]	62 Sm 150.36	63 Eu 151.96	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97				
**	90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np [237]	94 Pu [242]	95 Am [243]	96 Cm [247]	97 Bk [247]	98 Cf [251]	99 Es [252]	100 Fm [257]	101 Md [258]	102 No [259]	103 Lr [262]				